

Construction of DNA codes with R-constraint, RC-constraint and fixed GC content

Sujata Bansal¹ and Pramod Kumar Kewat¹

¹Department of Mathematics and Computing, Indian Institute of Technology (Indian School of Mines), Dhanbad, India 826004*

Abstract

DNA codes play a crucial role in DNA-based computation and storage, where combinatorial constraints are required to prevent unintended hybridization and to ensure thermodynamic stability. In this work, we investigate a family of reversible linear codes $\mathcal{C}_s$ over $\mathbb{F}_4$ and their duals $\mathcal{C}_s^\perp$ and provide a systematic construction of DNA codes satisfying multiple biologically motivated constraints. By analyzing the algebraic structure of $\mathcal{C}_s^\perp$, we explicitly classify its codewords and derive the complete weight enumerator and the GC-weight enumerator. Using these results, we construct several families of DNA codes over $\mathbb{F}_4$ with infinite cardinality that simultaneously satisfy the Hamming distance constraint together with the R-constraint, RC-constraint, fixed GC-content, or combinations thereof. The proposed framework provides a unified algebraic approach for designing large DNA codes with multiple constraints.

Key Words: Linear codes, DNA codes, Reversible codes.

1 Introduction

The use of DNA molecules for solving combinatorial and computational problems was first demonstrated by Adleman [Adl94], who successfully solved a seven-node instance of the directed Hamiltonian path problem using DNA strands. His approach utilized the Watson–Crick complementarity property of DNA and marked the first practical realization of computation at the molecular level. This breakthrough established DNA computing as a feasible paradigm for tackling computationally hard

*21dr0194@mc.iitism.ac.in, pramodk@iitism.ac.in

problems. Following this seminal work, Boneh et al. [BDL95] and, independently, Adleman et al. [Adl94] proposed molecular algorithms for breaking the Data Encryption Standard (DES), further highlighting the computational potential of DNA-based methods. Subsequent studies also indicated that DNA molecules can serve as a reliable medium for information storage [BGH⁺16, CGK12, MKK⁺03]. These developments led to a wide range of applications of DNA codes, including cryptanalysis, secure communication, and large-scale digital data storage using synthetic DNA (see [ARRW99, Lip95, SLM⁺96]).

Several combinatorial constraints are imposed on DNA codes to ensure reliable performance in molecular computing and storage applications [FLT⁺97, Kin03, LLCC02, MCC01]. A DNA code $\mathcal{C}$ is said to satisfy the Hamming distance constraint if $d_H(\mathbf{c}, \mathbf{c}') \geq d$ for all distinct $\mathbf{c}, \mathbf{c}' \in \mathcal{C}$ for some fixed integer d . It satisfies the reverse constraint (R-constraint) if $d_H(\mathbf{c}^R, \mathbf{c}') \geq d$ for every $\mathbf{c}, \mathbf{c}' \in \mathcal{C}$ and the reverse-complement constraint (RC-constraint) if $d_H(\mathbf{c}^{RC}, \mathbf{c}') \geq d$ for all $\mathbf{c}, \mathbf{c}' \in \mathcal{C}$, including the case $\mathbf{c} = \mathbf{c}'$. A DNA code $\mathcal{C}$ satisfies the fixed GC-content constraint if all its codewords have the same GC-weight. The first three constraints are designed to prevent undesirable hybridization between DNA strands, while the fixed GC-content constraint ensures uniform thermodynamic behavior, allowing parallel hybridization due to similar melting temperatures [SLM⁺96]. These constraints are fundamental in DNA-based computation and storage systems.

Gaborit and King [GK05] proposed algebraic constructions of DNA codes using linear and additive codes over $\mathbb{F}_4$, which initiated extensive research on constraint-based DNA code design. In particular, reversible codes over $\mathbb{F}_4$ have been widely used to construct DNA codes satisfying reverse and reverse-complement constraints [CCL26, KCL21, KŞU22, OYS17, PPY21]. In this work, we use a family of reversible linear codes over $\mathbb{F}_4$ introduced in [BK26] to construct several families of DNA codes satisfying multiple constraints simultaneously. Moreover, we determine the exact parameters of these families.

2 Preliminaries

2.1 DNA code over $\mathbb{F}_4$

For a vector $\mathbf{a} = (a_0, \dots, a_{n-1}) \in \mathbb{F}_4^n$, the reverse and reverse complement are defined as $\mathbf{a}^R = (a_{n-1}, \dots, a_0)$, and $\mathbf{a}^{RC} = (a_{n-1} + 1, \dots, a_0 + 1) = \mathbf{a}^R + \mathbf{1}_n$, respectively, where $\mathbf{1}_n$ denotes the all-one vector of length n over $\mathbb{F}_4$. In polynomial form each vector $\mathbf{a} = (a_0, \dots, a_{n-1}) \in \mathbb{F}_4^n$ can be represented as $a(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} \in \mathbb{F}_4[x]$. The reverse of $a(x)$ is given by $a(x)^R = x^{n-1}a(1/x)$, and its reverse complement is $a(x)^{RC} = a(x)^R + (1 + x + \dots + x^{n-1})$. Let C be a code of length n over $\mathbb{F}_4$. The Hamming

distance of C is defined by $d_H(C) = \min \{d_H(\mathbf{c}, \mathbf{c}') \mid \mathbf{c}, \mathbf{c}' \in C, \mathbf{c} \neq \mathbf{c}'\}$. If C is a vector subspace of $\mathbb{F}_4^n$, then it is called a linear code. A linear code of length n , dimension k and Hamming distance d is said to have parameters $[n, k, d]$. Such a code is called an optimal code if there exists no linear code with parameters $[n, k, d + 1]$. The dual of a linear code C is defined as $C^\perp = \{\mathbf{a} \in \mathbb{F}_4^n \mid \mathbf{a} \cdot \mathbf{c} = 0 \text{ for all } \mathbf{c} \in C\}$, where $\mathbf{a} \cdot \mathbf{c} = \sum_{i=0}^{n-1} a_i c_i$.

Let $\mathcal{A} = \{A, C, G, T\}$ denote the DNA alphabet consisting of four nucleotides. The Watson–Crick complement of a nucleotide is denoted by $\hat{\cdot}$, where $\hat{A} = T$, $\hat{T} = A$, $\hat{G} = C$, and $\hat{C} = G$. A non-empty subset of $\mathcal{A}^n$ is called a DNA code of length n . For a sequence $\mathbf{a} = (a_0, a_1, \dots, a_{n-1}) \in \mathcal{A}^n$, its reverse and reverse complement are defined as $\mathbf{a}^R = (a_{n-1}, a_{n-2}, \dots, a_0)$, and $\mathbf{a}^{RC} = (\hat{a}_{n-1}, \hat{a}_{n-2}, \dots, \hat{a}_0)$, respectively. The GC-weight of $\mathbf{a} \in \mathcal{A}^n$ is the number of components of $\mathbf{a}$ belonging to $\{G, C\}$. The Hamming distance between two sequences is the number of positions at which they differ.

Throughout this paper, we identify the DNA alphabet $\mathcal{A} = \{A, C, G, T\}$ with the elements of $\mathbb{F}_4 = \{0, \alpha, \alpha + 1, 1\}$ in the given order, where $\alpha^2 + \alpha + 1 = 0$, and hence view DNA codes over $\mathcal{A}$ as codes over the finite field $\mathbb{F}_4$. Consequently, $\{0, 1\} \simeq \{A, T\}$ and $\{\alpha, \alpha + 1\} \simeq \{C, G\}$. Under this identification, the Watson–Crick complement corresponds to the mapping $\hat{\beta} = \beta + 1$ for all $\beta \in \mathbb{F}_4$.

2.2 Weight enumerator of codes

Let C be a code of length n over $\mathbb{F}_4$. The complete weight enumerator and GC-weight enumerator of C are given by

$$CWE_C(x_0, x_1, x_\alpha, x_{\alpha+1}) = \sum_{\mathbf{c} \in C} x_0^{n_0(\mathbf{c})} x_1^{n_1(\mathbf{c})} x_\alpha^{n_\alpha(\mathbf{c})} x_{\alpha+1}^{n_{\alpha+1}(\mathbf{c})},$$

$$GCW_C(x, y) = \sum_{i=0}^n G_i x^{n-i} y^i,$$

where $n_\beta(\mathbf{c})$ denotes the number of occurrence of $\beta \in \mathbb{F}_4$ in $\mathbf{c}$ and G_i denotes the number of codewords in C having exactly i coordinates from $\{\alpha, \alpha + 1\}$ and $n - i$ coordinates from $\{0, 1\}$. It follows immediately that $GCW_C(x, y) = CWE_C(x, x, y, y)$.

3 A family of linear codes and its dual over $\mathbb{F}_4$

In this section, we recall a family of optimal linear codes over $\mathbb{F}_4$ and their duals, introduced in [BK26]. Let $s \geq 2$ be a fixed integer. For $1 \leq t \leq s - 1$ and $0 \leq l \leq 2^{s-t} - 1$, we define the polynomial $p_l^t(x) = x^{2^t l} + x^{2^t l + 2^{t-1}}$. Let $\Lambda_s^t = \{0, 1, \dots, 2^{s-t} - 1\}$. Then for $1 \leq t \leq s - 1$, define

$$\mathcal{A}_s^t = \left\{ \sum_{l \in \Gamma} p_l^t(x) \mid \Gamma \subseteq \Lambda_s^t, |\Gamma| = 2j, 0 \leq j \leq 2^{s-t-1} \right\},$$

where we consider $\sum_{l \in \emptyset} p_l^t(x) = 0$. Next, define the $\mathcal{F}_s^t = \mathcal{A}_s^t + \alpha \mathcal{A}_s^t = \{a(x) + \alpha b(x) \mid a(x), b(x) \in \mathcal{A}_s^t\}$, and $\mathcal{C}_s = \sum_{t=1}^{s-1} \mathcal{F}_s^t = \{f_1(x) + f_2(x) + \dots + f_{s-1}(x) \mid f_t(x) \in \mathcal{F}_s^t\}$. The parameters of $\mathcal{C}_s$ are given by the following result.

Theorem 1. ([BK26], Theorem 3.4) For every $s \geq 2$, $\mathcal{C}_s$ is a linear code over $\mathbb{F}_4$ with parameters $[2^s, 2^s - s - 1, 4]$.

Next, we consider the following $s + 1$ polynomials in $\mathbb{F}_4[x]$.

$$g_1(x) = \sum_{l=0}^{2^{s-1}-1} (x^{2l} + x^{2l+1}) = 1 + x + x^2 + \dots + x^{2^s-1}, \quad (1)$$

$$g_j(x) = \sum_{k=0}^{2^{j-2}-1} \left(\sum_{l=2k \cdot 2^{s-j}}^{(2k+1)2^{s-j}-1} (x^{2l} + x^{2l+1}) \right), \text{ where } 2 \leq j \leq s, \quad (2)$$

$$g_{s+1}(x) = \sum_{l=0}^{2^{s-2}-1} (x^{4l} + x^{4l+2}) = \sum_{l=0}^{2^{s-1}-1} x^{2l} = 1 + x^2 + \dots + x^{2^s-2}. \quad (3)$$

The dual code $\mathcal{C}_s^\perp$ is characterized as follows.

Theorem 2. ([BK26], Theorem 4.1) For every $s \geq 2$, the set $\mathcal{G} = \{g_j(x) \mid 1 \leq j \leq s + 1\}$ forms a basis of $\mathcal{C}_s^\perp$ over $\mathbb{F}_4$.

$$\mathcal{C}_s^\perp = \{a_1 g_1(x) + d(x) + a_{s+1} g_{s+1}(x) + \alpha(b_1 g_1(x) + d'(x) + b_{s+1} g_{s+1}(x)) \mid a_1, b_1, a_{s+1}, b_{s+1} \in \mathbb{F}_2 \text{ and } d(x), d'(x) \in D_s\}, \quad (4)$$

where $D_s = \left\{ \sum_{j=2}^s a_j g_j(x) \mid a_j \in \mathbb{F}_2 \right\}$.

From [BK26], we have the following results.

Theorem 3. Let $p(x) \in D_s \setminus \{0\}$. Then $p(x)$ can be expressed as $p(x) = \sum_{l \in P} (x^{2l} + x^{2l+1})$, where $P \subseteq \Lambda_s^1$ and $|P| = 2^{s-2}$.

Theorem 4. Let $p(x) = \sum_{l \in P} (x^{2l} + x^{2l+1})$ and $q(x) = \sum_{l \in Q} (x^{2l} + x^{2l+1})$ be two distinct elements of D_s , where $P, Q \subseteq \Lambda_s^1$ and $|P| = |Q| = 2^{s-2}$. Then $|P \cap Q| = 2^{s-3}$.

Finally, by [BK26], the parameters of the dual code are as follows.

Theorem 5. $\mathcal{C}_s^\perp$ is a linear code over $\mathbb{F}_4$ with parameters $[2^s, s + 1, 2^{s-1}]$.

4 DNA codes from $\mathcal{C}_s$ and $\mathcal{C}_s^\perp$ satisfying multiple constraints

In this section, we construct DNA codes over $\mathbb{F}_4$ that satisfy multiple constraints by using the family of linear codes $\mathcal{C}_s$ and their duals $\mathcal{C}_s^\perp$, introduced in Section 3. Throughout this work, all codes under consideration satisfy the Hamming distance constraint. Therefore, we focus on additional biologically motivated constraints such as the reverse and reverse-complement

constraints. Let $\mathcal{C}$ be a code over $\mathbb{F}_4$ such that the reverse of every codeword also belongs to $\mathcal{C}$. We partition $\mathcal{C}$ into two disjoint subsets $\mathcal{C}'$ and $\mathcal{C}''$, where $\mathcal{C}'$ is the set of self-reversible codewords and $\mathcal{C}''$ is the set of non-self-reversible codewords. The set $\mathcal{C}''$ can be further decomposed into two disjoint subsets $\tilde{\mathcal{C}}''$ and $\tilde{\mathcal{C}}''^R$, where $\tilde{\mathcal{C}}'' \subseteq \mathcal{C}''$, and $\tilde{\mathcal{C}}''^R = \{\mathbf{c}^R \mid \mathbf{c} \in \tilde{\mathcal{C}}''\}$. This decomposition is not unique, and the cardinality of $\mathcal{C}$ satisfies $|\mathcal{C}| = |\mathcal{C}'| + 2|\tilde{\mathcal{C}}''|$. Moreover, each of the sets $\tilde{\mathcal{C}}''$ and $\tilde{\mathcal{C}}''^R$ individually satisfies the R-constraint together with the Hamming distance constraint.

An analogous decomposition applies when $\mathbf{c}^{RC} \in \mathcal{C}$ for all $\mathbf{c} \in \mathcal{C}$. In this case, the code $\mathcal{C}$ can be partitioned into self reverse-complement codewords and pairs of mutually reverse-complement codewords. Consequently, one can construct subcodes that independently satisfy the RC-constraint along with the Hamming distance constraint.

We now consider a specific family of linear codes $\mathcal{C}_s$ and their duals over $\mathbb{F}_4$, where $s \geq 2$. We show that the reverse and reverse-complement of each codeword in $\mathcal{C}_s$ (respectively, $\mathcal{C}_s^\perp$) also belong to $\mathcal{C}_s$ (respectively, $\mathcal{C}_s^\perp$).

Lemma 1. ([BK26], Theorem 6.1) *For every $c(x) \in \mathcal{C}_s$, $c(x)^R \in \mathcal{C}_s$.*

Lemma 2. *For every $c(x) \in \mathcal{C}_s^\perp$, we have $c(x)^R \in \mathcal{C}_s^\perp$.*

Proof. By Theorem 2, the set $\mathcal{G} = \{g_j(x) \mid 1 \leq j \leq s+1\}$ forms a basis of $\mathcal{C}_s^\perp$ over $\mathbb{F}_4$. Thus, any $c(x) \in \mathcal{C}_s^\perp$ can be expressed as $c(x) = \sum_{j=1}^{s+1} \beta_j g_j(x)$, where $\beta_j \in \mathbb{F}_4$. Taking reverses, we obtain $c(x)^R = \sum_{j=1}^{s+1} \beta_j g_j(x)^R$. Since $g_1(x)^R = g_1(x)$, and $g_j(x)^R = g_j(x) + g_1(x)$ for $2 \leq j \leq s+1$, it follows that $g_j(x)^R \in \mathcal{C}_s^\perp$ for all j . Hence, $c(x)^R \in \mathcal{C}_s^\perp$. This completes the proof. $\square$

Lemma 3. *For every $c(x) \in \mathcal{C}_s$, its reverse-complement $c(x)^{RC} \in \mathcal{C}_s$.*

Proof. For $c(x) \in \mathcal{C}_s$, we have $c(x)^{RC} = c(x)^R + (1 + x + \dots + x^{2^s-1})$. By Lemma 1, $c(x)^R \in \mathcal{C}_s$, and since $1 + x + \dots + x^{2^s-1} \in \mathcal{C}_s$, the claim follows. $\square$

Since $g_1(x) = 1 + x + \dots + x^{2^s-1} \in \mathcal{C}_s^\perp$, thus we have the following result.

Lemma 4. *For every $c(x) \in \mathcal{C}_s^\perp$, its reverse-complement $c(x)^{RC} \in \mathcal{C}_s^\perp$.*

Thus, both $\mathcal{C}_s$ and $\mathcal{C}_s^\perp$ are closed under reversal and reverse-complement, and hence can be used to construct DNA codes satisfying various constraints.

Using the representation in Eq. (4), the codewords of $\mathcal{C}_s^\perp$ are classified into four cases based on $(d(x), d'(x))$, where $d(x)$ and $d'(x)$ are elements of D_s . Each case is further divided into 16 subcases according to the choice of $(a_1, b_1, a_{s+1}, b_{s+1}) \in \mathbb{F}_2^4$. As a result, $\mathcal{C}_s^\perp$ is partitioned into 64 mutually disjoint subcodes. A detailed description of this partition can be found in [BK26]. For each subcode, we compute the GC-weight of its codewords and determine the corresponding complete weight enumerator. The complete weight enumerator of $\mathcal{C}_s^\perp$ is obtained as the sum of the complete weight

enumerators of all 64 subcodes. We also identify, for each subcode, whether its codewords satisfy $c(x)^R = c(x)$, $c(x)^{RC} = c(x)$, or neither. Let n_R , n_{RC} , and n_O denote the numbers of codewords satisfying $c(x)^R = c(x)$, $c(x)^{RC} = c(x)$, and neither condition, respectively. The results are summarized in four tables, corresponding to the four cases. Due to page limitations, we present only the final outcomes in tabular form; detailed proofs will be included in the final version. For all cases listed in Tables 2 and 4, every codeword has GC-weight 2^{s-1} .

Table 1: Codewords of $\mathcal{C}_s^\perp$ when $d(x) = d'(x) = 0$

Case	$c(x)$	CWE	GCW ($c(x)$)	(n_R, n_{RC}, n_O)
1	0	$x_0^{2^s}$	0	(1, 0, 0)
2	$g_1(x)$	$x_1^{2^s}$	0	(1, 0, 0)
3	$g_{s+1}(x)$	$x_0^{2^{s-1}} x_1^{2^{s-1}}$	0	(0, 1, 0)
4	$g_1(x) + g_{s+1}(x)$	$x_0^{2^{s-1}} x_1^{2^{s-1}}$	0	(0, 1, 0)
5	$\alpha g_1(x)$	$x_\alpha^{2^s}$	2^s	(1, 0, 0)
6	$(1 + \alpha)g_1(x)$	$x_{\alpha+1}^{2^s}$	2^s	(1, 0, 0)
7	$g_{s+1}(x) + \alpha g_1(x)$	$x_\alpha^{2^{s-1}} x_{\alpha+1}^{2^{s-1}}$	2^s	(0, 1, 0)
8	$(1 + \alpha)g_1(x) + g_{s+1}(x)$	$x_\alpha^{2^{s-1}} x_{\alpha+1}^{2^{s-1}}$	2^s	(0, 1, 0)
9	$\alpha g_{s+1}(x)$	$x_0^{2^{s-1}} x_\alpha^{2^{s-1}}$	2^{s-1}	(0, 0, 1)
10	$g_1(x) + \alpha g_{s+1}(x)$	$x_1^{2^{s-1}} x_{\alpha+1}^{2^{s-1}}$	2^{s-1}	(0, 0, 1)
11	$(1 + \alpha)g_{s+1}(x)$	$x_0^{2^{s-1}} x_{\alpha+1}^{2^{s-1}}$	2^{s-1}	(0, 0, 1)
12	$g_1(x) + (1 + \alpha)g_{s+1}(x)$	$x_1^{2^{s-1}} x_\alpha^{2^{s-1}}$	2^{s-1}	(0, 0, 1)
13	$\alpha (g_1(x) + g_{s+1}(x))$	$x_0^{2^{s-1}} x_\alpha^{2^{s-1}}$	2^{s-1}	(0, 0, 1)
14	$(1 + \alpha)g_1(x) + \alpha g_{s+1}(x)$	$x_1^{2^{s-1}} x_{\alpha+1}^{2^{s-1}}$	2^{s-1}	(0, 0, 1)
15	$\alpha g_1(x) + (1 + \alpha)g_{s+1}(x)$	$x_1^{2^{s-1}} x_\alpha^{2^{s-1}}$	2^{s-1}	(0, 0, 1)
16	$(1 + \alpha)(g_1(x) + g_{s+1}(x))$	$x_0^{2^{s-1}} x_{\alpha+1}^{2^{s-1}}$	2^{s-1}	(0, 0, 1)

From Tables 1-4, we obtain the following result.

Theorem 6. For $s \geq 2$, there are 2^{2s} codewords in $\mathcal{C}_s^\perp$ satisfying $c(x)^R = c(x)$ and 2^{2s} codewords satisfying $c(x)^{RC} = c(x)$. Furthermore, in each case, among these 2^{2s} codewords, 2^s codewords have GC-weight 0, $2^{2s} - 2^{s+1}$ codewords have GC-weight 2^{s-1} , and 2^s codewords have GC-weight 2^s .

Complete weight and GC-weight enumerator of $\mathcal{C}_s$ and $\mathcal{C}_s^\perp$:

Using the data summarized in Tables 1-4, the CWE of $\mathcal{C}_s^\perp$ is given by

$$\begin{aligned}
CWE_{\mathcal{C}_s^\perp}(x_0, x_1, x_\alpha, x_{\alpha+1}) &= x_0^{2^s} + x_1^{2^s} + x_\alpha^{2^s} + x_{\alpha+1}^{2^s} + (2^{s+1} - 2) \\
&\quad \left(x_0^{2^{s-1}} x_1^{2^{s-1}} + x_0^{2^{s-1}} x_\alpha^{2^{s-1}} + x_0^{2^{s-1}} x_{\alpha+1}^{2^{s-1}} + x_1^{2^{s-1}} x_\alpha^{2^{s-1}} + x_1^{2^{s-1}} x_{\alpha+1}^{2^{s-1}} \right. \\
&\quad \left. + x_\alpha^{2^{s-1}} x_{\alpha+1}^{2^{s-1}} \right) + 8(2^{s-1} - 1)(2^s - 1)x_0^{2^{s-2}} x_1^{2^{s-2}} x_\alpha^{2^{s-2}} x_{\alpha+1}^{2^{s-2}}.
\end{aligned}$$

Table 2: Codewords of $\mathcal{C}_s^\perp$ when $d(x) = 0$, $d'(x) \neq 0$.

Here, $\mathcal{U}_1 = (2^{s-1} - 1)x_0^{2^{s-1}}x_\alpha^{2^{s-1}}$, $\mathcal{U}_2 = (2^{s-1} - 1)x_1^{2^{s-1}}x_{\alpha+1}^{2^{s-1}}$,
 $\mathcal{U}_3 = (2^{s-1} - 1)x_0^{2^{s-2}}x_1^{2^{s-2}}x_\alpha^{2^{s-2}}x_{\alpha+1}^{2^{s-2}}$.

Case	$c(x)$	CWE	(n_R, n_{RC}, n_O)
1	$\alpha d'(x)$	$\mathcal{U}_1$	$(2^{s-2} - 1, 0, 2^{s-2})$
2	$g_1(x) + \alpha d'(x)$	$\mathcal{U}_2$	$(2^{s-2} - 1, 0, 2^{s-2})$
3	$g_{s+1}(x) + \alpha d'(x)$	$\mathcal{U}_3$	$(0, 2^{s-2} - 1, 2^{s-2})$
4	$g_1(x) + g_{s+1}(x) + \alpha d'(x)$	$\mathcal{U}_3$	$(0, 2^{s-2} - 1, 2^{s-2})$
5	$\alpha(g_{s+1}(x) + d'(x))$	$\mathcal{U}_1$	$(2^{s-2}, 0, 2^{s-2} - 1)$
6	$g_1(x) + \alpha g_{s+1}(x) + \alpha d'(x)$	$\mathcal{U}_2$	$(2^{s-2}, 0, 2^{s-2} - 1)$
7	$(1 + \alpha)g_{s+1}(x) + \alpha d'(x)$	$\mathcal{U}_3$	$(0, 2^{s-2}, 2^{s-2} - 1)$
8	$g_1(x) + (1 + \alpha)g_{s+1}(x) + \alpha d'(x)$	$\mathcal{U}_3$	$(0, 2^{s-2}, 2^{s-2} - 1)$
9	$\alpha(g_1(x) + d'(x))$	$\mathcal{U}_1$	$(2^{s-2} - 1, 0, 2^{s-2})$
10	$(1 + \alpha)g_1(x) + \alpha d'(x)$	$\mathcal{U}_2$	$(2^{s-2} - 1, 0, 2^{s-2})$
11	$\alpha g_1(x) + g_{s+1}(x) + \alpha d'(x)$	$\mathcal{U}_3$	$(0, 2^{s-2} - 1, 2^{s-2})$
12	$(1 + \alpha)g_1(x) + g_{s+1}(x) + \alpha d'(x)$	$\mathcal{U}_3$	$(0, 2^{s-2} - 1, 2^{s-2})$
13	$\alpha(g_1(x) + g_{s+1}(x)) + \alpha d'(x)$	$\mathcal{U}_1$	$(2^{s-2}, 0, 2^{s-2} - 1)$
14	$(1 + \alpha)g_1(x) + \alpha(g_{s+1}(x) + d'(x))$	$\mathcal{U}_2$	$(2^{s-2}, 0, 2^{s-2} - 1)$
15	$\alpha g_1(x) + (1 + \alpha)g_{s+1}(x) + \alpha d'(x)$	$\mathcal{U}_3$	$(0, 2^{s-2}, 2^{s-2} - 1)$
16	$(1 + \alpha)(g_1(x) + g_{s+1}(x)) + \alpha d'(x)$	$\mathcal{U}_3$	$(0, 2^{s-2}, 2^{s-2} - 1)$

Consequently, the GC-weight enumerator of $\mathcal{C}_s^\perp$ is

$$GCW_{\mathcal{C}_s^\perp}(x, y) = 2^{s+1}x^{2^s} + 2^{s+1}y^{2^s} + 2^{s+2}(2^s - 1)x^{2^{s-1}}y^{2^{s-1}}.$$

By the MacWilliams identity ([HP10], Eq. 7.13), the CWE of $\mathcal{C}_s$ is

$$CWE_{\mathcal{C}_s}(x_0, x_1, x_\alpha, x_{\alpha+1}) = \frac{1}{|\mathcal{C}_s^\perp|} CWE_{\mathcal{C}_s^\perp}\left(x_0 + x_1 + x_\alpha + x_{\alpha+1},\right. \\ \left. x_0 + x_1 - x_\alpha - x_{\alpha+1}, x_0 - x_1 - x_\alpha + x_{\alpha+1}, x_0 - x_1 + x_\alpha - x_{\alpha+1}\right),$$

and hence, the GC-weight enumerator of $\mathcal{C}_s$ is

$$GCW_{\mathcal{C}_s}(x, y) = 2^{2^s-2s-2} \left((x+y)^{2^s} + (x-y)^{2^s} + (2^{s+1} - 2) \right. \\ \left. (x+y)^{2^{s-1}}(x-y)^{2^{s-1}} \right).$$

After simplification, we can write $GCW_{\mathcal{C}_s}(x, y)$ as $\sum_{i=0}^{2^s} G_i x^{2^s-i} y^i$. Then $G_i = 2^{2^s-2s-2} \left((1 + (-1)^i) \binom{2^s}{i} + (2^{s+1} - 2) \sum_{j=0}^i ((-1)^{i-j} \binom{2^{s-1}}{j} \binom{2^{s-1}}{i-j}) \right)$ for $0 \leq i \leq 2^{s-1}$, and $G_i = 2^{2^s-2s-2} \left((1 + (-1)^i) \binom{2^s}{i} + (2^{s+1} - 2) \sum_{j=0}^{2^s-i} ((-1)^{2^{s-1}-j} \binom{2^{s-1}}{2^{s-1}-j} \binom{2^{s-1}}{i-2^{s-1}+j}) \right)$ for $2^{s-1} + 1 \leq i \leq 2^s$.

Table 3: Codewords of $\mathcal{C}_s^\perp$ when $d(x) \neq 0$, $d'(x) = 0$.

Here, $\mathcal{V}_1 = (2^{s-1} - 1)x_0^{2^{s-1}}x_1^{2^{s-1}}$, $\mathcal{V}_2 = (2^{s-1} - 1)x_\alpha^{2^{s-1}}x_{\alpha+1}^{2^{s-1}}$,
 $\mathcal{V}_3 = (2^{s-1} - 1)x_0^{2^{s-2}}x_1^{2^{s-2}}x_\alpha^{2^{s-2}}x_{\alpha+1}^{2^{s-2}}$.

Case	$c(x)$	CWE	GCW ($c(x)$)	(n_R, n_{RC}, n_O)
1	$d(x)$	$\mathcal{V}_1$	0	$(2^{s-2} - 1, 2^{s-2}, 0)$
2	$g_{s+1}(x) + d(x)$	$\mathcal{V}_1$	0	$(2^{s-2}, 2^{s-2} - 1, 0)$
3	$g_1(x) + d(x)$	$\mathcal{V}_1$	0	$(2^{s-2} - 1, 2^{s-2}, 0)$
4	$g_1(x) + g_{s+1}(x) + d(x)$	$\mathcal{V}_1$	0	$(2^{s-2}, 2^{s-2} - 1, 0)$
5	$\alpha g_1(x) + d(x)$	$\mathcal{V}_2$	2^s	$(2^{s-2} - 1, 2^{s-2}, 0)$
6	$g_{s+1}(x) + \alpha g_1(x) + d(x)$	$\mathcal{V}_2$	2^s	$(2^{s-2}, 2^{s-2} - 1, 0)$
7	$(1 + \alpha)g_1(x) + d(x)$	$\mathcal{V}_2$	2^s	$(2^{s-2} - 1, 2^{s-2}, 0)$
8	$(1 + \alpha)g_1(x) + g_{s+1}(x) + d(x)$	$\mathcal{V}_2$	2^s	$(2^{s-2}, 2^{s-2} - 1, 0)$
9	$\alpha g_{s+1}(x) + d(x)$	$\mathcal{V}_3$	2^{s-1}	$(0, 0, 2^{s-1} - 1)$
10	$(1 + \alpha)g_{s+1}(x) + d(x)$	$\mathcal{V}_3$	2^{s-1}	$(0, 0, 2^{s-1} - 1)$
11	$g_1(x) + \alpha g_{s+1}(x) + d(x)$	$\mathcal{V}_3$	2^{s-1}	$(0, 0, 2^{s-1} - 1)$
12	$g_1(x) + (1 + \alpha)g_{s+1}(x) + d(x)$	$\mathcal{V}_3$	2^{s-1}	$(0, 0, 2^{s-1} - 1)$
13	$\alpha(g_1(x) + g_{s+1}(x)) + d(x)$	$\mathcal{V}_3$	2^{s-1}	$(0, 0, 2^{s-1} - 1)$
14	$(1 + \alpha)g_{s+1}(x) + \alpha g_1(x) + d(x)$	$\mathcal{V}_3$	2^{s-1}	$(0, 0, 2^{s-1} - 1)$
15	$(1 + \alpha)g_1(x) + \alpha g_{s+1}(x) + d(x)$	$\mathcal{V}_3$	2^{s-1}	$(0, 0, 2^{s-1} - 1)$
16	$(1 + \alpha)(g_1(x) + g_{s+1}(x)) + d(x)$	$\mathcal{V}_3$	2^{s-1}	$(0, 0, 2^{s-1} - 1)$

DNA codes from $\mathcal{C}_s^\perp$ with R-constraint or RC-constraint:

$\mathcal{C}_s^\perp$ is a linear code over $\mathbb{F}_4$ with length 2^s , size 2^{2s+2} , and Hamming distance 2^{s-1} . By Theorem 6, exactly 2^{2s} codewords satisfy $c(x)^R = c(x)$. Hence,

$$|\mathcal{C}_s^{\perp R}| = |\mathcal{C}_s^{\perp R}| = \frac{|\mathcal{C}_s^\perp| - 2|\mathcal{C}_s^{\perp'}|}{2} = \frac{2^{2s+2} - 2^{2s}}{2} = 3 \cdot 2^{2s-1}.$$

Therefore, for $s \geq 2$, the sets $\mathcal{C}_s^{\perp R}$ and $\mathcal{C}_s^{\perp R}$ form DNA codes of length 2^s over $\mathbb{F}_4$ with size $3 \cdot 2^{2s-1}$, Hamming distance at least 2^{s-1} and satisfying the reverse constraint. An analogous construction yields DNA codes satisfying the reverse-complement constraint with the same parameters.

DNA codes with fixed GC-content constraint from $\mathcal{C}_s^\perp$:

From the GC-weight enumerator of $\mathcal{C}_s^\perp$, we define two subcodes $\mathcal{C}_{s,1}^\perp$ and $\mathcal{C}_{s,2}^\perp$ of $\mathcal{C}_s^\perp$. The set $\mathcal{C}_{s,1}^\perp$ consists codewords with GC-weight 2^{s-1} and $\mathcal{C}_{s,2}^\perp$ consists codewords with GC-weight 2^s . Then $\mathcal{C}_{s,1}^\perp$ is a DNA code of length 2^s , size $2^{s+2}(2^s - 1)$, fixed GC-content 2^{s-1} , and Hamming distance at least 2^{s-1} , while $\mathcal{C}_{s,2}^\perp$ is a DNA code of length 2^s , size 2^{s+1} , fixed GC-content 2^s , and Hamming distance at least 2^{s-1} .

DNA codes with R(or RC)-constraint and fixed GC-content from $\mathcal{C}_s^\perp$:

Table 4: Codewords of $\mathcal{C}_s^\perp$ when $d(x) \neq 0$, $d'(x) \neq 0$.

Here, $\mathcal{W}_1 = (2^{s-1} - 1)x_0^{2^{s-1}}x_{\alpha+1}^{2^{s-1}} + (2^{s-1} - 1)(2^{s-1} - 2)x_0^{2^{s-2}}x_1^{2^{s-2}}x_\alpha^{2^{s-2}}x_{\alpha+1}^{2^{s-2}}$, $\mathcal{W}_2 = (2^{s-1} - 1)^2x_0^{2^{s-2}}x_1^{2^{s-2}}x_\alpha^{2^{s-2}}x_{\alpha+1}^{2^{s-2}}$, $\mathcal{W}_3 = (2^{s-1} - 1)x_1^{2^{s-1}}x_\alpha^{2^{s-1}} + (2^{s-1} - 1)(2^{s-1} - 2)x_0^{2^{s-2}}x_1^{2^{s-2}}x_\alpha^{2^{s-2}}x_{\alpha+1}^{2^{s-2}}$,
 $w_1 = ((2^{s-2} - 1)^2, 2^{s-2}(2^{s-2} - 1), 2^{s-2}(2^{s-1} - 1))$,
 $w_2 = (2^{s-2}(2^{s-2} - 1), (2^{s-2} - 1)^2, 2^{s-2}(2^{s-1} - 1))$,
 $w_3 = (2^{s-2}(2^{s-2} - 1), (2^{s-2})^2, (2^{s-2} - 1)(2^{s-1} - 1))$,
 $w_4 = ((2^{s-2})^2, 2^{s-2}(2^{s-2} - 1), (2^{s-2} - 1)(2^{s-1} - 1))$

Case	$c(x)$	CWE	(n_R, n_{RC}, n_O)
1	$d(x) + \alpha d'(x)$	$\mathcal{W}_1$	w_1
2	$g_{s+1}(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_2$	w_2
3	$g_1(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_3$	w_1
4	$g_1(x) + g_{s+1}(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_2$	w_2
5	$\alpha g_{s+1}(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_2$	w_3
6	$(1 + \alpha)g_{s+1}(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_1$	w_4
7	$g_1(x) + \alpha g_{s+1}(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_2$	w_3
8	$g_1(x) + (1 + \alpha)g_{s+1}(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_3$	w_4
9	$\alpha g_1(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_3$	w_1
10	$g_{s+1}(x) + \alpha g_1(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_2$	w_2
11	$(1 + \alpha)g_1(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_1$	w_1
12	$(1 + \alpha)g_1(x) + g_{s+1}(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_2$	w_2
13	$\alpha(g_1(x) + g_{s+1}(x)) + d(x) + \alpha d'(x)$	$\mathcal{W}_2$	w_3
14	$(1 + \alpha)g_{s+1}(x) + \alpha g_1(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_3$	w_4
15	$(1 + \alpha)g_1(x) + \alpha g_{s+1}(x) + d(x) + \alpha d'(x)$	$\mathcal{W}_2$	w_3
16	$(1 + \alpha)(g_1(x) + g_{s+1}(x)) + d(x) + \alpha d'(x)$	$\mathcal{W}_1$	w_4

By Theorem 6, the GC-weight enumerator of the set of self-reversible codewords in $\mathcal{C}_s^\perp$ is given by

$$GCW_{\mathcal{C}_s^\perp} = 2^s x^{2^s} + (2^{2s} - 2^{s+1}) x^{2^{s-1}} y^{2^{s-1}} + 2^s y^{2^s}.$$

Consequently, the GC-weight enumerator of the set $\mathcal{C}_s^{\perp''}$ is

$$GCW_{\mathcal{C}_s^{\perp''}}(x, y) = 2^s x^{2^s} + (3 \cdot 2^{2s} - 2^{s+1}) x^{2^{s-1}} y^{2^{s-1}} + 2^s y^{2^s}.$$

Hence,

$$GCW_{\mathcal{C}_s^{\perp''}}(x, y) = 2^{s-1} x^{2^s} + 2^s (3 \cdot 2^{s-1} - 1) x^{2^{s-1}} y^{2^{s-1}} + 2^{s-1} y^{2^s}.$$

Therefore, for $s \geq 2$, we obtain a DNA code of length 2^s over $\mathbb{F}_4$ with size $2^s (3 \cdot 2^{s-1} - 1)$ that satisfies both the reverse constraint and a fixed GC-content equal to 2^{s-1} , as well as a DNA code of size 2^{s-1} satisfying the reverse constraint with fixed GC-content 2^s .

By an analogous argument, the same conclusions hold for DNA codes satisfying the reverse-complement constraint together with fixed GC-content.

We illustrate the above construction with the following example.

Example 1. By Theorem 1, the code $\mathcal{C}_4$ is a linear code over $\mathbb{F}_4$ with parameters $[16, 11, 4]$. We write $\mathcal{C}_4$ as a union of three disjoint subsets $\mathcal{C}_4 = \mathcal{C}'_4 \cup \tilde{\mathcal{C}}''_4 \cup \tilde{\mathcal{C}}''^R_4$, where $\mathcal{C}'_4$ denotes the set of self-reversible codewords. Using Magma computations, we obtain $|\mathcal{C}'_4| = 16,384$. Since $\mathcal{C}_4 = 4^{11}$, it follows that $|\tilde{\mathcal{C}}''_4| = |\tilde{\mathcal{C}}''^R_4| = \frac{4^{11} - 16,384}{2} = 2,088,960$. Thus, both $\tilde{\mathcal{C}}''_4$ and $\tilde{\mathcal{C}}''^R_4$ form DNA codes of length 16 over $\mathbb{F}_4$ satisfying the reverse constraint with Hamming distance at least 4. Moreover, the GC-weight enumerator of $\mathcal{C}_4$ is given by $GCW_{\mathcal{C}_4}(x, y) = 2048x^{16} + 286720x^{12}y^4 + 917504x^{10}y^6 + 1781760x^8y^8 + 917504x^6y^{10} + 286720x^4y^{12} + 2048y^{16}$. Thus, $\mathcal{C}_4$ yields several DNA codes of length 16 over $\mathbb{F}_4$ with fixed GC-contents 4, 6, 8, 10, 12, and 16, whose sizes are given by the coefficients of $GCW_{\mathcal{C}_4}(x, y)$.

Table 5 summarizes the parameters of the families of DNA codes over $\mathbb{F}_4$ of length 2^s and Hamming distance at least 2^{s-1} constructed from $\mathcal{C}_s^\perp$, each satisfying different constraints. Moreover, identical parameters are obtained for the corresponding families that satisfy the reverse-complement constraint in place of the reverse constraint.

Table 5: Families of DNA codes constructed from $\mathcal{C}_s^\perp$

Family	Constraints satisfied	Size
$\mathcal{C}_s^{\tilde{\perp}''}$	Reverse	$3 \cdot 2^{2s-1}$
$\mathcal{C}_{s,1}^\perp$	Fixed GC-content 2^{s-1}	$2^{s+2}(2^s - 1)$
$\mathcal{C}_{s,2}^\perp$	Fixed GC-content 2^s	2^{s+1}
$\mathcal{C}_s^{\tilde{\perp}''} \cap \mathcal{C}_{s,1}^\perp$	Reverse + fixed GC-content 2^{s-1}	$2^s(3 \cdot 2^{s-1} - 1)$
$\mathcal{C}_s^{\tilde{\perp}''} \cap \mathcal{C}_{s,2}^\perp$	Reverse + fixed GC-content 2^s	2^{s-1}

5 Conclusion

In this paper, we investigated a family of optimal linear codes $\mathcal{C}_s$ over $\mathbb{F}_4$ and their duals $\mathcal{C}_s^\perp$ to construct DNA codes satisfying multiple constraints. We showed that both $\mathcal{C}_s$ and $\mathcal{C}_s^\perp$ are closed under the reverse and reverse-complement operations, which enables systematic constructions of DNA codes with these constraints. By analyzing the structure of $\mathcal{C}_s^\perp$, we obtained explicit classifications of codewords and derived the complete weight enumerator and GC-weight enumerator. These results led to several infinite families of DNA codes over $\mathbb{F}_4$ satisfying multiple constraints, with explicitly determined parameters.

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